

GPU-Accelerated Skin Disease Detection using RAPIDS and Deep Learning

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ABSTRACT

This project proposes a GPU-accelerated system for skin disease detection using deep learning and RAPIDS libraries. The system integrates Convolutional Neural Networks (CNNs) with RAPIDS tools such as cuDF and cuML to optimize data preprocessing, training, and inference. The goal is to develop a scalable and efficient medical image classification pipeline capable of real-time performance.

1 INTRODUCTION AND MOTIVATION

Skin diseases are among the most common global health concerns, yet early diagnosis remains limited due to lack of access to dermatological expertise. Artificial intelligence, particularly deep learning, has demonstrated strong potential in automating skin disease classification using image data.

However, traditional machine learning pipelines rely heavily on CPU-based data processing, resulting in slow training times and limited scalability. This project aims to overcome these limitations by integrating GPU-accelerated data science tools from RAPIDS, enabling faster preprocessing, model training, and deployment.

The primary goal is to build an efficient, scalable skin disease detection system that leverages both deep learning and high-performance GPU computing.

2 PROBLEM STATEMENT AND OBJECTIVES

Problem: Existing skin disease detection systems face challenges such as:

- High computational cost of training deep learning models
- Slow data preprocessing pipelines
- Limited scalability for large datasets

Objectives:

- Develop a CNN-based skin disease classification model
- Integrate RAPIDS (cuDF, cuML) for GPU-accelerated data processing
- Reduce training and preprocessing time
- Build a scalable and deployable system

3 METHODOLOGY

The proposed system consists of the following components:

1. Data Collection and Preprocessing

A dataset of labeled skin disease images will be used. RAPIDS cuDF will handle data loading, cleaning, and preprocessing on GPU.

2. Data Pipeline Optimization

cuML will be used for preprocessing operations such as scaling and feature handling, ensuring faster execution compared to CPU pipelines.

3. Model Development

A Convolutional Neural Network (CNN) will be implemented using TensorFlow or PyTorch for image classification.

4. Training and Optimization

Training will be performed on GPU, leveraging RAPIDS for efficient data flow. Hyperparameter tuning will improve performance.

5. Evaluation

Model performance will be evaluated using accuracy, precision, recall, and F1-score.

6. Deployment

The final model will be deployed as a web or API-based application for real-time predictions.

4 EXPECTED RESULTS

The project is expected to achieve:

- High accuracy in skin disease classification
- Reduced training and preprocessing time using RAPIDS
- Improved scalability for large datasets

A comparison between traditional and RAPIDS-based pipelines will demonstrate performance improvements.

5 IMPACT

This project contributes to:

- Faster AI development in healthcare
- Improved accessibility to early disease detection tools
- Advancement of GPU-accelerated data science techniques

The system can support telemedicine and assist healthcare professionals in diagnosis.

6 PROJECT TIMELINE

- Month 1: Dataset preparation and preprocessing using cuDF
- Month 2: Model development and training
- Month 3: Optimization using RAPIDS and evaluation
- Month 4: Deployment and documentation

7 RESOURCES REQUIRED

- GPU-enabled system
- RAPIDS libraries (cuDF, cuML)
- Deep learning frameworks (TensorFlow/PyTorch)

8 CONCLUSION

This project presents a scalable and efficient approach to skin disease detection by combining deep learning with RAPIDS GPU acceleration. The proposed system reduces computational overhead while maintaining high accuracy, making it suitable for real-world healthcare applications.